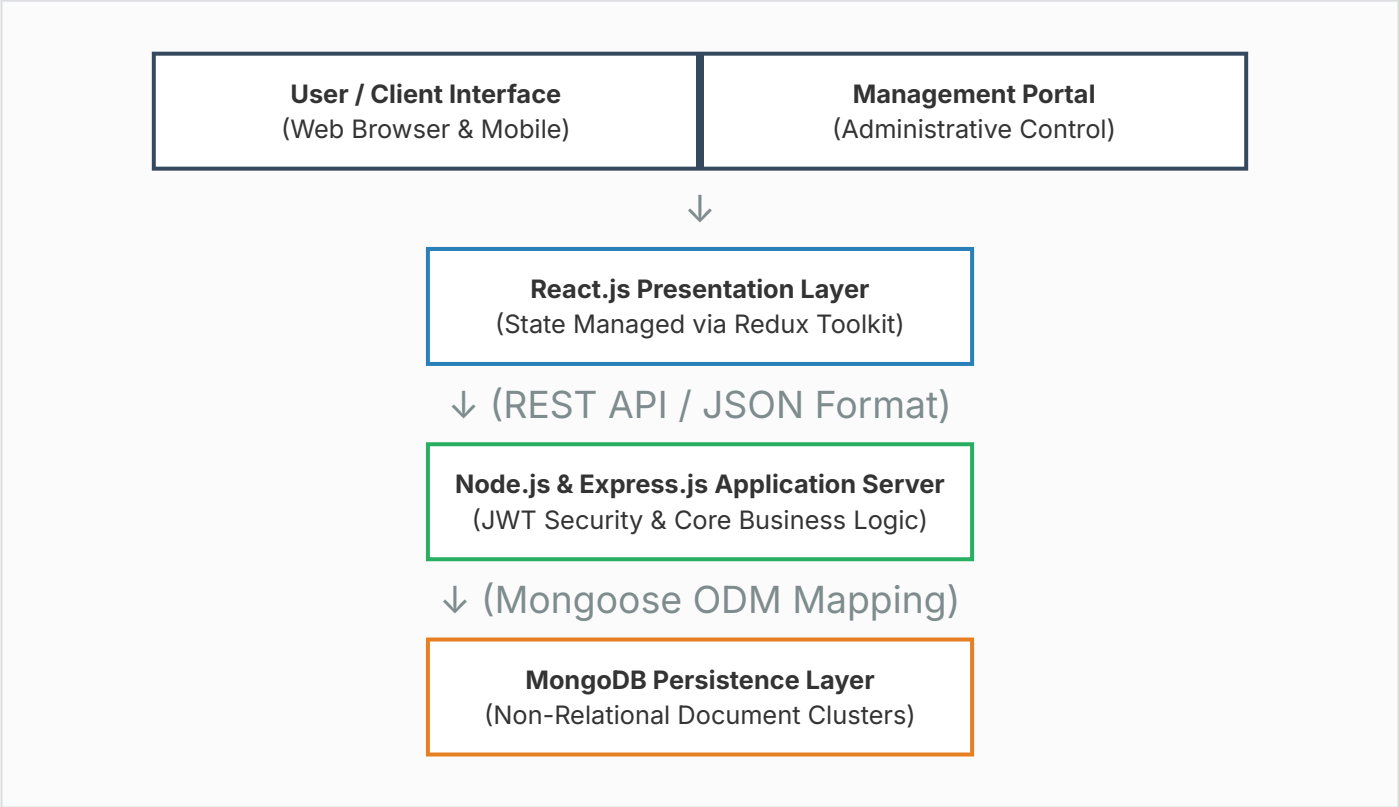


PROJECT DESIGN PHASE-II

Technology Stack & System Architecture

Date	18 June 2026	Project Name	ShopEz (MERN E-Commerce)
Team ID	--	Maximum Marks	4 Marks

1. Technical Architecture (MERN Stack)



2. Components & Technology Stack

Component	Description	Technology
Client Interface	Dynamic Single Page Application (SPA) providing tailored views for both consumers and store leads.	React.js, Tailwind CSS
Backend Server	The central processing hub handling incoming REST requests, user validation, and system routing.	Node.js & Express.js
Primary Database	Fully managed cloud-native document store for persistent storage of customer and catalog metadata.	MongoDB Atlas
Authentication	Robust mechanism for secure user sessions, utilizing cryptographically signed tokens and hashed secrets.	JWT & Bcrypt.js
State Management	Centralized handling of global application states like checkout progress and active user sessions.	React Context API

Component	Description	Technology
Infrastructure	Scalable deployment environments encompassing local development and cloud hosting platforms.	Local / Vercel / Heroku

3. Module Design & Data Architecture

Module	Functional Description	Key Database Fields
Auth Module	Orchestrates user on-boarding, credential verification, and secure access token issuance.	email , password (hash) , role
Product Module	Governs the digital storefront, including multi-tier categorization, inventory updates, and search.	name , price , category , stock
Order Module	Manages the lifecycle of a sale, from initial placement through to delivery history tracking.	userID , items[] , total , status

4. API Interface Design (REST)

Endpoint	Method	Purpose
/api/users/login	POST	Validates user artifacts and provides an authorized JWT.
/api/products	GET	Retrieves the full product set or filters based on category.
/api/orders/place	POST	Submits new transaction metadata to the database cluster.
/api/admin/product	POST/PUT	Facilitates administrative catalog CRUD operations.

5. Design Considerations

Elastic Scalability: By employing a NoSQL (MongoDB) foundation, the system accommodates rapid data growth and horizontal scaling across server clusters. **Platform Security:** Sensitive API routes are fortified with JWT-based middleware, ensuring that user passwords are encrypted rather than stored as plain text. **Execution Performance:** React's optimized Virtual DOM implementation facilitates near-instantaneous UI refreshes during complex sorting, product filtering, and shopping basket updates.